

BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE421 / EEE465: Computer Networks — Final Examination

Semester: Spring 2024 — Set: B (v2) — Full Marks: 70 — Duration: 2 Hours 5 Minutes

COMPREHENSIVE SOLUTION AND COMPLETE ANSWER KEY

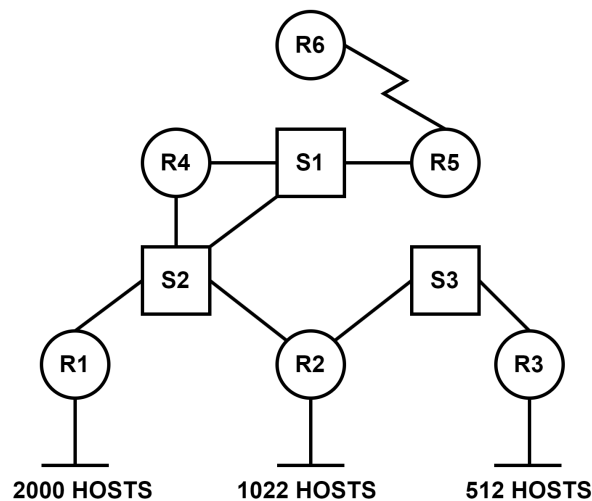
SECTION A: MANDATORY QUESTIONS (40 MARKS)

Question 1: Hierarchical VLSM Subnetting

[CO3]

3 + 12 = 15 Marks

Problem Statement: Mahalabia Inc. needs to subnet the existing network (1.2.128.0/17) into smaller sub-networks as given in the topology below.



I. Determine the maximum number of subnets you can create from the above-mentioned root network. [3 Marks]

Solution:

- The root network is 1.2.128.0/17.
- Total host bits available in the root network = $32 - 17 = 15$ bits.
- In IPv4 network design, the smallest standard subnet capable of supporting usable host devices (such as point-to-point router WAN links) is a /30 subnet (providing 2 usable host IPs, 1 network address, and 1 broadcast address).
- Subnet bits borrowed to create /30 subnets = $30 - 17 = 13$ bits.
- Maximum number of /30 subnets:

$$\text{Max Subnets} = 2^{13} = \mathbf{8192} \text{ subnets}$$

(Note: If considering theoretical host routes of prefix /32, $2^{32-17} = 2^{15} = 32,768$ individual addresses).

Answer: 8,192 subnets (at /30).

II. Efficiently calculate the network address for all the sub-networks and draw the hierarchical tree. [12 Marks]

0.1 Topology Requirement Analysis & Subnet Identification: In IPv4 network architecture, subnets (broadcast domains) are demarcated by router Layer 3 interfaces. Analyzing the given multi-router campus topology:

- **LAN Subnets (End Devices):**

1. **R1 LAN:** Requires **2000 Hosts**.
2. **R2 LAN:** Requires **1022 Hosts**.
3. **R3 LAN:** Requires **512 Hosts**.

- **Inter-Router Subnets (Switches & WAN Links):**

4. **Switch S2 Subnet:** Connects router interfaces of **R1, R2, and R4** $\Rightarrow$ **3 Hosts** (3 router IP interfaces).
5. **Switch S1 Subnet:** Connects router interfaces of **R4 and R5** $\Rightarrow$ **2 Hosts** (2 router IP interfaces).
6. **Switch S3 Subnet:** Connects router interfaces of **R2 and R3** $\Rightarrow$ **2 Hosts** (2 router IP interfaces).
7. **WAN Link (R5–R6):** Point-to-point serial link between **R5 and R6** $\Rightarrow$ **2 Hosts** (2 router IP interfaces).

(Architectural Note: If the diagonal inter-switch link between S1 and S2 is treated as a unified Layer 2 broadcast domain, S1 and S2 would merge into a single transit subnet connecting 4 router interfaces R1, R2, R4, R5 requiring $H = 4 \Rightarrow$ /29. Operating S1 and S2 as distinct routed subnets provides 7 dedicated, cleanly separated subnets as derived below).

0.2 Step-by-Step Derivations (Following the 6-Step Method):

1. Subnet 1: R1 LAN (2000 Hosts — Largest)

- **Step 1 (Capacity H_2):** $H_2 = H + 2 = 2000 + 2 = 2002$
- **Step 2 (Host Bits H_b):** $H_b = \lceil \log_2(2002) \rceil = 11$ bits ($2^{10} = 1024 < 2002 \leq 2^{11} = 2048$)
- **Step 3 (Prefix P):** $P = 32 - 11 = /21$
- **Step 4 (Allocated IPs):** $2^{11} = 2048$ IPs (Block size in 3rd octet = $2048/256 = 8$)
- **Step 5 (Addresses):**
 - Network Address: 1.2.128.0/21
 - Usable Host Range: 1.2.128.1 to 1.2.135.254
 - Directed Broadcast Address: 1.2.135.255
 - Next Subnet Base IP: $1.2.(128 + 8).0 = 1.2.136.0$
- **Step 6 (Subnet Mask):** $255.255.(256 - 8).0 = 255.255.248.0$

2. Subnet 2: R2 LAN (1022 Hosts)

- **Step 1 (Capacity H_2):** $H_2 = H + 2 = 1022 + 2 = 1024$
- **Step 2 (Host Bits H_b):** $H_b = \lceil \log_2(1024) \rceil = 10$ bits ($2^9 = 512 < 1024 \leq 2^{10} = 1024$)
- **Step 3 (Prefix P):** $P = 32 - 10 = /22$
- **Step 4 (Allocated IPs):** $2^{10} = 1024$ IPs (Block size in 3rd octet = $1024/256 = 4$)
- **Step 5 (Addresses):**
 - Network Address: 1.2.136.0/22
 - Usable Host Range: 1.2.136.1 to 1.2.139.254
 - Directed Broadcast Address: 1.2.139.255
 - Next Subnet Base IP: 1.2.(136 + 4).0 = 1.2.140.0
- **Step 6 (Subnet Mask):** 255.255.(256 - 4).0 = **255.255.252.0**

3. Subnet 3: R3 LAN (512 Hosts)

- **Step 1 (Capacity H_2):** $H_2 = H + 2 = 512 + 2 = 514$
- **Step 2 (Host Bits H_b):** $H_b = \lceil \log_2(514) \rceil = 10$ bits ($2^9 = 512 < 514 \leq 2^{10} = 1024$)
- **Step 3 (Prefix P):** $P = 32 - 10 = /22$
- **Step 4 (Allocated IPs):** $2^{10} = 1024$ IPs (Block size in 3rd octet = 4)
- **Step 5 (Addresses):**
 - Network Address: 1.2.140.0/22
 - Usable Host Range: 1.2.140.1 to 1.2.143.254
 - Directed Broadcast Address: 1.2.143.255
 - Next Subnet Base IP: 1.2.(140 + 4).0 = 1.2.144.0
- **Step 6 (Subnet Mask):** 255.255.(256 - 4).0 = **255.255.252.0**
- *(Alternative Note: If 512 was loosely intended as the power-of-two block allocation 2^9 , then $H_b = 9 \implies /23$ with 510 usable hosts, spanning 1.2.140.0/23 and freeing 1.2.142.0 for subsequent subnets).*

4. Subnet 4: Switch S2 Subnet (3 Hosts: R1, R2, R4)

- **Step 1 (Capacity H_2):** $H_2 = 3 + 2 = 5$
- **Step 2 (Host Bits H_b):** $H_b = \lceil \log_2(5) \rceil = 3$ bits ($2^2 = 4 < 5 \leq 2^3 = 8$)
- **Step 3 (Prefix P):** $P = 32 - 3 = /29$
- **Step 4 (Allocated IPs):** $2^3 = 8$ IPs
- **Step 5 (Addresses):**
 - Network Address: 1.2.144.0/29
 - Usable Host Range: 1.2.144.1 to 1.2.144.6

- Directed Broadcast Address: 1.2.144.7
- Next Subnet Base IP: 1.2.144.0 + 8 = 1.2.144.8

- **Step 6 (Subnet Mask):** 255.255.255.(256 – 8) = **255.255.255.248**

5. Subnet 5: Switch S1 Subnet (2 Hosts: R4, R5)

- **Step 1 (Capacity H_2):** $H_2 = 2 + 2 = 4$
- **Step 2 (Host Bits H_b):** $H_b = \lceil \log_2(4) \rceil = 2$ bits ($2^2 = 4$)
- **Step 3 (Prefix P):** $P = 32 - 2 = /30$
- **Step 4 (Allocated IPs):** $2^2 = 4$ IPs
- **Step 5 (Addresses):**
 - Network Address: 1.2.144.8/30
 - Usable Host Range: 1.2.144.9 to 1.2.144.10
 - Directed Broadcast Address: 1.2.144.11
 - Next Subnet Base IP: 1.2.144.8 + 4 = 1.2.144.12
- **Step 6 (Subnet Mask):** 255.255.255.(256 – 4) = **255.255.255.252**

6. Subnet 6: Switch S3 Subnet (2 Hosts: R2, R3)

- **Step 1 (Capacity H_2):** $H_2 = 2 + 2 = 4 \implies H_b = 2$ bits
- **Step 3 (Prefix P):** $P = 32 - 2 = /30$, Total Allocated IPs = 4
- **Step 5 (Addresses):**
 - Network Address: 1.2.144.12/30
 - Usable Host Range: 1.2.144.13 to 1.2.144.14
 - Directed Broadcast Address: 1.2.144.15
 - Next Subnet Base IP: 1.2.144.12 + 4 = 1.2.144.16
- **Step 6 (Subnet Mask):** **255.255.255.252**

7. Subnet 7: WAN Serial Link R5–R6 (2 Hosts)

- **Step 1 (Capacity H_2):** $H_2 = 2 + 2 = 4 \implies H_b = 2$ bits
- **Step 3 (Prefix P):** $P = 32 - 2 = /30$, Total Allocated IPs = 4
- **Step 5 (Addresses):**
 - Network Address: 1.2.144.16/30
 - Usable Host Range: 1.2.144.17 to 1.2.144.18
 - Directed Broadcast Address: 1.2.144.19
 - Next Available Base IP: 1.2.144.20
- **Step 6 (Subnet Mask):** **255.255.255.252**

0.3 Complete Master VLSM Allocation Table:

Subnet Name	Req. Hosts (H)	H_b	Prefix	Network IP	Usable Host Range	Subnet Mask
R1 LAN	2000	11	/21	1.2.128.0	1.2.128.1 – 1.2.135.254	255.255.248.0
R2 LAN	1022	10	/22	1.2.136.0	1.2.136.1 – 1.2.139.254	255.255.252.0
R3 LAN	512	10	/22	1.2.140.0	1.2.140.1 – 1.2.143.254	255.255.252.0
Switch S2 (R1, R2, R4)	3	3	/29	1.2.144.0	1.2.144.1 – 1.2.144.6	255.255.255.24
Switch S1 (R4, R5)	2	2	/30	1.2.144.8	1.2.144.9 – 1.2.144.10	255.255.255.25
Switch S3 (R2, R3)	2	2	/30	1.2.144.12	1.2.144.13 – 1.2.144.14	255.255.255.25
WAN Link (R5–R6)	2	2	/30	1.2.144.16	1.2.144.17 – 1.2.144.18	255.255.255.25

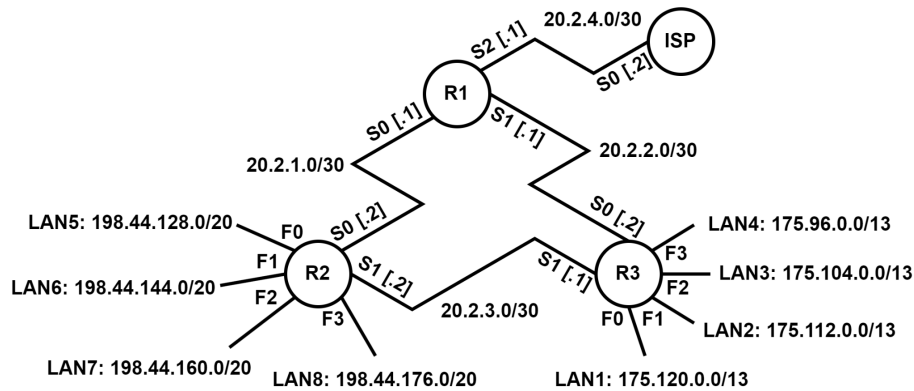
0.4 Hierarchical Subnet Tree:

```

1.2.128.0/17 (Root Network, 32,768 IPs: 1.2.128.0 to 1.2.255.255)
|-- 1.2.128.0/18 (First Branch: 16,384 IPs: 1.2.128.0 to 1.2.191.255)
|   |-- 1.2.128.0/21 [R1 LAN: 2000 Hosts] (2,048 IPs)
|   |-- 1.2.136.0/22 [R2 LAN: 1022 Hosts] (1,024 IPs)
|   |-- 1.2.140.0/22 [R3 LAN: 512 Hosts] (1,024 IPs)
|   '-- 1.2.144.0/22 (Transit & Core Blocks)
|       |-- 1.2.144.0/29 [Switch S2: R1, R2, R4] (8 IPs)
|       |-- 1.2.144.8/30 [Switch S1: R4, R5] (4 IPs)
|       |-- 1.2.144.12/30 [Switch S3: R2, R3] (4 IPs)
|       |-- 1.2.144.16/30 [WAN Link: R5--R6] (4 IPs)
|       '-- 1.2.144.20/30 ... 1.2.147.255 (Free Unallocated Pool)
'-- 1.2.192.0/18 (Second Branch: 16,384 IPs Reserved for Future Campus Expansion)

```

Question 2: Route Summarization & Floating Static Route [CO3] 6 + 4 = 10 Marks



I. Calculate the summarized network of all R3 LANs and configure a summary static route in R1. [6 Marks]

Solution:

1. **List R3 LANs:**

- LAN4: 175.96.0.0/13
- LAN3: 175.104.0.0/13
- LAN2: 175.112.0.0/13
- LAN1: 175.120.0.0/13

2. **Binary Expansion of Second Octet:**

$$96_{10} = 01100000_2$$

$$104_{10} = 01101000_2$$

$$112_{10} = 01110000_2$$

$$120_{10} = 01111000_2$$

The first 3 bits of the second octet (011) are identical across all four networks.

3. Total common network bits = 8 (1st octet) + 3 (2nd octet) = /11.

4. Summarized Network Address: 175.96.0.0/11

5. Subnet Mask for /11: 255.224.0.0

6. **Configuration Command on Router R1:** R1 reaches R3 directly via interface S1 (link 20.2.2.0/30, R3 is S0 [.2] → next-hop IP 20.2.2.2):

```
R1(config)# ip route 175.96.0.0 255.224.0.0 20.2.2.2
```

(or specifying exit interface: R1(config)# ip route 175.96.0.0 255.224.0.0 S1).

II. Floating Static Route on R1 to LAN2 of R3 via R2 with AD = 50. [4 Marks]

Solution:

- Destination Network: LAN2 of R3 = 175.112.0.0/13 (Subnet Mask: 255.248.0.0).
- Outbound exit interface towards Router R2 is S0 (link 20.2.1.0/30).
- Next-hop IP on Router R2 is 20.2.1.2.
- Administrative Distance required: **50**.
- Cisco IOS Command on R1:

R1(config)# ip route 175.112.0.0 255.248.0.0 S0 50

(or with next-hop IP: R1(config)# ip route 175.112.0.0 255.248.0.0 S0 20.2.1.2 50).

Question 3: IPv4 Fragmentation Analysis

[CO3]

2 + 2 + 1 = 5 Marks

Given:

- Original datagram total length = 4560 bytes (includes 20-byte IP header).
- Outbound serial link MTU = 380 bytes.

I. Calculate the number of fragments created. [2 Marks]

- Total original payload data = $4560 - 20 = 4540$ bytes.
- Maximum payload per fragment $\leq 380 - 20 = 360$ bytes.
- Since $360/8 = 45$ is an exact multiple of 8, each intermediate fragment carries exactly **360 bytes of data**.
- Number of full 360-byte fragments = $\lfloor 4540/360 \rfloor = 12$ fragments.
- Data accounted for by first 12 fragments = $12 \times 360 = 4320$ bytes.
- Remaining data for the 13th (last) fragment = $4540 - 4320 = 220$ bytes.
- **Total number of fragments** = $12 + 1 = 13$ fragments.

II. Calculate the fragment offset of the 4th fragment. [2 Marks]

- Using the standard fragment offset formula from the networking study notes:

$$\text{Fragment Offset}_k = \frac{(k - 1) \times \text{Payload}}{8}$$

where k is the fragment index ($k = 4$) and Payload = 360 bytes is the data payload per full fragment.

- Substituting $k = 4$ and Payload = 360:

$$\text{Offset}_4 = \frac{(4 - 1) \times 360}{8} = \frac{3 \times 360}{8} = 3 \times 45 = \mathbf{135}$$

- (Verification: The starting byte in the original unfragmented payload is $3 \times 360 = 1080$ bytes, spanning bytes 1080 to 1439).

Answer: 135.

III. State the MF bit of the 5th fragment. [1 Mark]

- Since there are 13 fragments, the 5th fragment is an intermediate fragment ($5 < 13$).
- Therefore, More Fragments flag: **MF = 1**.

Problem Statement: Dipu tries to share Zahin's DHCP server located two floors down in another company's network. However, Dipu's devices cannot receive any IP configuration from the DHCP server. Identify the problem and explain the solution.

Solution:

1. Identification of the Problem [3 Marks]:

- When Dipu's client devices boot up, they broadcast **DHCPDISCOVER** messages with destination IP 255.255.255.255 and destination MAC FF:FF:FF:FF:FF:FF.
- Zahin's DHCP server resides on a different network across one or more intermediate routers.
- By architectural design, **routers do not forward Layer 2 or Layer 3 broadcast frames** across broadcast domain boundaries.
- Therefore, the DHCPDISCOVER messages are blocked at Dipu's local router and never reach Zahin's DHCP server.

2. Solution [3 Marks]:

- Configure Dipu's router interface as a **DHCP Relay Agent** (using the Cisco IOS command `ip helper-address <Zahin_DHCP_Server_IP>` on the router interface connected to Dipu's LAN).
- The router intercepts the broadcast DHCP packets, converts them into unicast UDP packets, sets the `giaddr` field to the router's local interface IP, and forwards them directly to Zahin's DHCP server.
- Zahin's DHCP server uses the `giaddr` to select the correct IP address pool for Dipu's subnet and unicasts the reply back to the router, which forwards it to Dipu's devices.

Question: Dipu can only afford a single public IP address. So, the ISP allows only one device to access the Internet. State if this statement is true or false, and explain your answer.

Solution:

- **The statement is FALSE.**
- **Technical Explanation:**
 - Even with only a single public IP address, all twenty devices in Dipu's startup company can simultaneously access the Internet using **Port Address Translation (PAT)**, also known as **NAT Overload**.
 - PAT allows multiple private hosts (using RFC 1918 addresses like 192.168.1.0/24) to share a single public IP address by mapping each connection to a unique Layer 4 source port number (TCP or UDP).
 - Since there are approximately 64,512 usable dynamic port numbers per IP, the router can maintain thousands of concurrent sessions for multiple devices at the same time.

SECTION B: ANY TWO OUT OF THREE (12 MARKS)

(All questions solved below for complete coverage)

Question 6: Distance Vector Periodic Routing Updates

[CO3]

6 Marks

Question: Refer to the figure in Q2. Assume all routers run Distance Vector Routing. Through which interfaces will Router R2 send routing updates periodically? Explain your answer.

Solution:

- In Distance Vector routing (e.g., RIPv2), routers periodically advertise their entire routing table out of **all active interfaces participating in the routing protocol process**.
- Looking at Router R2:
 - Interface S0 connects to Router R1 (WAN link 20.2.1.0/30).
 - Interface S1 connects to Router R3 (WAN link 20.2.3.0/30).
 - Interfaces F0, F1, F2, F3 connect to LAN5, LAN6, LAN7, and LAN8.
- **Routing Adjacencies:** Router R2 must send dynamic routing updates across its serial interfaces S0 (to R1) and S1 (to R3) so that adjacent routers learn about R2's LANs and transit paths.
- On the LAN interfaces (F0--F3), by default RIP sends updates, but in professional network practice, these are configured as **passive interfaces** (`passive-interface`) to prevent advertising routing information onto user networks where no routers exist.
- Therefore, active periodic routing exchange occurs through interfaces S0 and S1.

Question 7: EUI-64 MAC Extraction & IPv6 Classification [CO3] 4 + 2 = 6 Marks

Given IPv6 Address: FE80:0:0:B0B:980:FF:FE00::

I. Identify the MAC address of your device. [4 Marks]

Solution:

1. The 128-bit IPv6 address expands to:

fe80:0000:0000:0b0b : 0980:00ff:fe00:0000

2. The lower 64 bits form the EUI-64 Interface Identifier:

Interface ID = 0980:00ff:fe00:0000

3. In EUI-64 generation:

- The reserved delimiter bytes FF:FE were inserted between the 3rd and 4th bytes of the 48-bit MAC address.
- Removing FF:FE yields the 6 bytes: 09 : 80 : 00 : 00 : 00 : 00.
- The 7th bit (Universal/Local bit) of the first byte was inverted during generation.

4. Reversing the 7th bit inversion on the first byte (09):

$$09_{16} = 0000100\mathbf{1}_2 \xrightarrow{\text{Invert bit 7}} 000010\mathbf{11}_2 = 0B_{16}$$

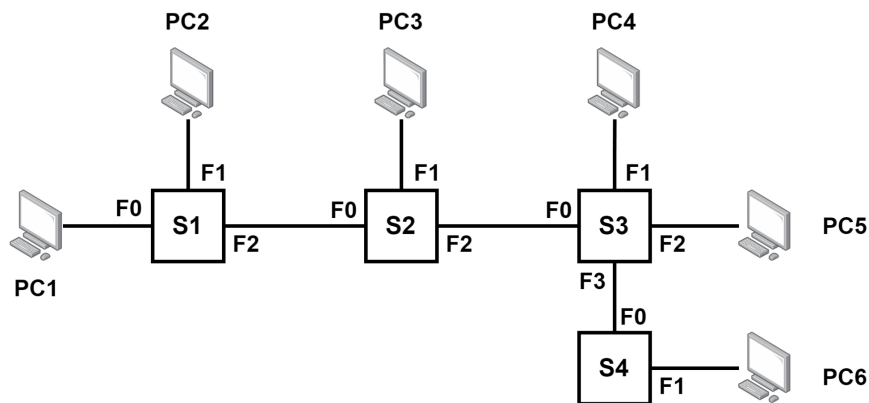
5. Therefore, the original 48-bit MAC address is:

$$0B : 80 : 00 : 00 : 00 : 00 \quad (\text{or } 0b:80:00:00:00:00)$$

II. Identify the type of IPv6 address. [2 Marks]

- The address prefix is FE80::/10.
- **Answer: Link-Local Unicast Address.**

Question 8: Switch Forwarding Behavior (Unicasting vs. Flooding) [CO2] 4 + 2 = 6 Marks



Initial MAC Tables:

- S2 contains: PC3 on F1, PC5 on F2.
- S3 contains: PC3 on F0, PC5 on F2.
- PC4 sends frames to PC3 and PC2. None of them replies.

I. How sending these two frames by switch S3 is different [4 Marks]:

1. Frame to PC3 (Known Unicast):

- Switch S3 already has an entry for PC3 pointing to port F0.
- Therefore, S3 performs **selective unicasting**: it forwards the frame solely out of port F0 towards S2.

2. Frame to PC2 (Unknown Unicast):

- S3 has no entry for PC2 in its MAC address table.
- Therefore, S3 performs **flooding**: it broadcasts the frame out of all active ports except the incoming port: F0 (towards S2), F2 (towards PC5), and F3 (towards S4).

II. Current state of MAC tables of S3 and S2 [2 Marks]:

- When PC4 transmits, S3 receives the frame on port F1 and learns: **PC4** → F1.
- The frame exits S3 on F0 and enters S2 on port F2. S2 learns: **PC4** → F2.

Switch S3 MAC Table

Device / MAC	Port
PC3	F0
PC5	F2
PC4	F1

Switch S2 MAC Table

Device / MAC	Port
PC3	F1
PC5	F2
PC4	F2

SECTION C: ANY THREE OUT OF FIVE (18 MARKS)

(All questions solved below for complete coverage)

Question 9: Denial-of-Service via Ping (ICMP Flood / Smurf) [CO2] 2 + 4 = 6 Marks

I. Name of the Attack [2 Marks]: ICMP Flood Attack (or Smurf Attack / Ping Flood DoS).

II. Process Explanation [4 Marks]:

- In an **ICMP Flood**, the attacker (or a distributed botnet) overwhelms the target web server by sending an enormous flood of ICMP Echo Request packets faster than the server can handle.
- In a **Smurf Attack**, the attacker broadcasts ICMP Echo Requests with the source address spoofed as the victim's web server IP to an amplification network. Hundreds of devices simultaneously send ICMP Echo Replies back to the victim.
- The victim's network bandwidth becomes saturated, and its operating system CPU/memory resources are consumed responding to ICMP traffic, rendering the server completely unresponsive to legitimate incoming HTTP requests.

Question 10: ARP Request Forwarding & Host Next Steps [CO2] 2 + 4 = 6 Marks

Context: Using the figure in Q8, PC1 sends an ARP request for PC5's MAC address.

I. How switches S1, S2, S3 forward this request [2 Marks]:

- An ARP Request is a broadcast frame (Destination MAC: FF:FF:FF:FF:FF:FF).
- **S1:** Receives frame on F0, learns PC1 on F0, and **floods** it out of F1 (to PC2) and F2 (to S2).
- **S2:** Receives frame on F0, learns PC1 on F0, and **floods** it out of F1 (to PC3) and F2 (to S3).
- **S3:** Receives frame on F0, learns PC1 on F0, and **floods** it out of F1 (to PC4), F2 (to PC5), and F3 (to S4).

II. Two functions PC1 will perform after receiving the ARP reply [4 Marks]:

1. **Cache the MAC Address:** PC1 updates its local **ARP cache** by storing the IP-to-MAC mapping of PC5.
2. **Encapsulate and Transmit Data:** PC1 dequeues the pending IP packet, encapsulates it in an Ethernet frame with Destination MAC set to PC5's MAC address, and transmits it onto the link.

Question 11: IPv6 over IPv4 Encapsulation (Tunneling) [CO2] 4 + 2 = 6 Marks

I. Name of the Process [2 Marks]: IPv6 Tunneling (e.g., 6in4, 6to4, or GRE Tunneling).

II. Why this is done [4 Marks]:

- During the ongoing global migration to IPv6, many intermediate transit networks and Internet Service Providers (ISPs) remain IPv4-only.

- Encapsulation allows two isolated IPv6 edge networks to exchange traffic across an IPv4-only core network.
- The entry border router encapsulates the entire IPv6 packet inside an IPv4 packet (Protocol type 41), which IPv4 routers route normally. The exit border router strips the IPv4 header and delivers the original IPv6 packet to the destination.

Question 12: MAC Classification & Flat vs. Hierarchical Addressing [CO2] 2 + 4 = 6 Marks

Given MAC Address: 98:CC:12:23:40:BB

I. Unicast or Multicast [2 Marks]:

- Look at the first octet: 98. In binary: $98_{16} = 10011000_2$.
- The Least Significant Bit (b_0) is **0** (or second hex digit '8' is even).
- **Answer: Unicast MAC Address.**

II. Why MAC is flat rather than hierarchical like IPv4 [4 Marks]:

- An IPv4 address is **hierarchical** (Network ID + Subnet ID + Host ID), allowing routers to summarize thousands of destinations into a single route entry based on geographic and network topology.
- In contrast, a MAC address is **flat**: it is permanently tied to hardware (burned-in) and carries **no topological location information**.
- If a device moves to another network, its MAC address remains identical. Switches cannot summarize MAC addresses, which is why MAC addressing cannot scale globally across the Internet.

Question 13: Topology Change Handling in Distance Vector [CO3] 6 Marks

Solution: While Link State triggers an immediate LSP flood, Distance Vector handles topology changes through:

1. **Failure Detection via Timers:** If periodic routing updates from a neighbor cease, an invalid/timeout timer (e.g., 180 seconds in RIP) expires, and the route is marked unreachable (metric = 16).
2. **Triggered Updates:** To speed up convergence, the detecting router immediately sends an update message without waiting for the regular 30-second broadcast timer.
3. **Loop Prevention Mechanisms:** Routers employ **Split Horizon** (do not advertise a route back out of the interface it was learned on) and **Poison Reverse** (explicitly advertising the broken route with metric $\infty = 16$) to halt count-to-infinity loops.
4. **Flush Timer:** After an additional flush period (e.g., 240 seconds), the unreachable route is purged from the routing table.

— END OF EXAMINATION SOLUTION —